Philosophy of Emergence

Session 2
07.06.2024

Emergence in complex systems

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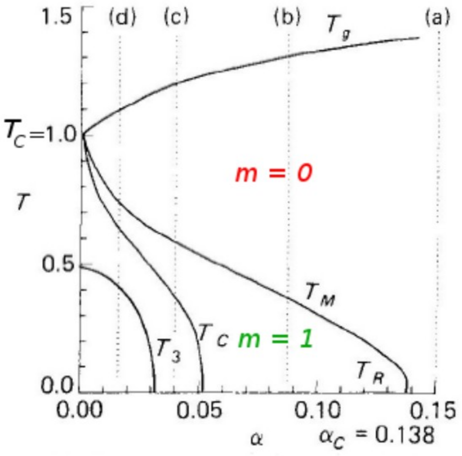
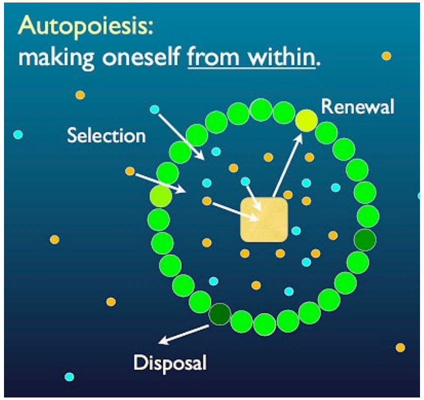
- Notions of emergence involved have weaker requirements
- Sometimes dismissed as trivial emergence, or emergence in an uninteresting sense...
- … which is a shame because it does not help clarify (and classify) the different concepts of emergence used in science

Emergence in complex systems

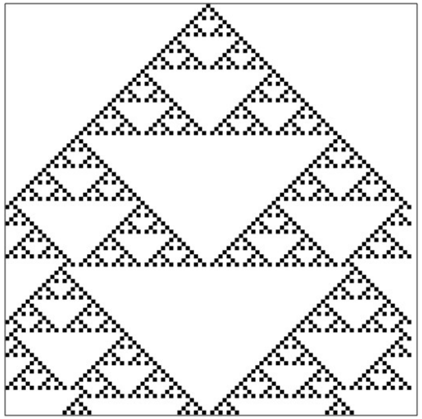
- "Emergent systems":



Systems that exhibit phase transitions



Autopoietic systems



Cellular automaton patterns

...

Images: Brunel (2004), http://www.whatlifeis.info, https://plato.stanford.edu/

Emergence in complex systems

- “Emergent systems”:
  - *What do they have in common?*
  - *In what sense are they emergent?*

Emergence in complex systems

- Bedau’s “weak emergence”
- The underdetermination of weak emergence
- Towards a typology of emergent phenomena

Weak emergence

Weak Emergence

Author(s): Mark A. Bedau

Source: *Philosophical Perspectives*, 1997, Vol. 11, Mind, Causation, and World (1997), pp. 375-399

Published by: Ridgeview Publishing Company

Stable URL: https://www.jstor.org/stable/2216138

Starts with a criticism of strong emergence, especially the downward causation thesis

- Downward causation is mysterious
- Downward causation is scientifically irrelevant

Weak emergence

Supervenient emergentism – 4 theses:

1. *Emergence*. When aggregates of material particles come to an adequate complexity level, authentically new properties emerge and characterize these systems.

2. *Irreducibility of emergent properties*. Emergent properties are not reducible to their “base conditions”, i.e., the underlying conditions they emerge from.

3. *Supervenience of emergents on their base conditions*. Emergent properties supervene on their base conditions.

4. *Downward causation*. Emergent properties possess causal powers to influence phenomena at the level they emerged from.

Weak emergence

A viable concept of emergence must meet 3 goals:

- Metaphysically innocent
- Consistent with materialism
- Scientifically useful

Weak emergence is present in all complex systems and meets these goals

Weak emergence

Definition of weak emergence

- *S*: system composed of different parts
- Microstates = states of *S*'s parts
- Macrostates = structural properties of *S*, constituted wholly out of the parts' microstates
- D: dynamics that governs the time evolution of *S*'s microstates

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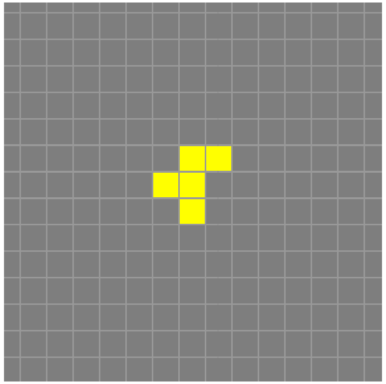
Important: the need to use a simulation does not depend on our knowledge
(≠ epistemological)

Weak emergence

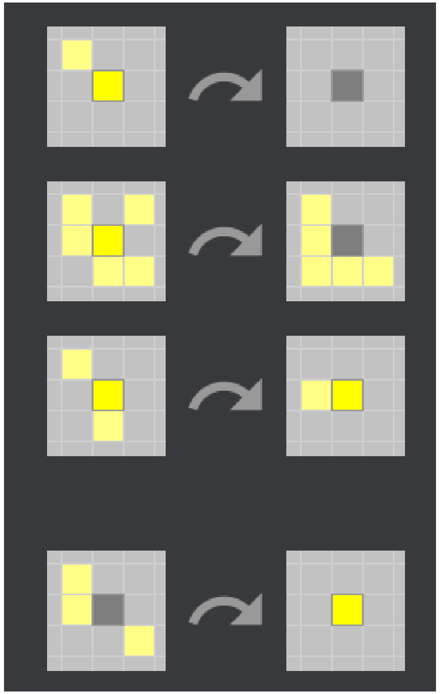
Example 1: the Game of Life (Conway, 1970)

Example of a weakly emergent macrostate in Life:

the R-pentomino's bounded growth



Birth-death rule



Images: https://playgameoflife.com/

Weak emergence

Example 2: Packard's model of evolving life

- Artificial agents that consume resources, reproduce and die
- Agents have a set of genes that determine their sensorimotor behavior
- Genes are transmitted to the next generation during reproduction
- Genes mutate during reproduction; mutation rate μ

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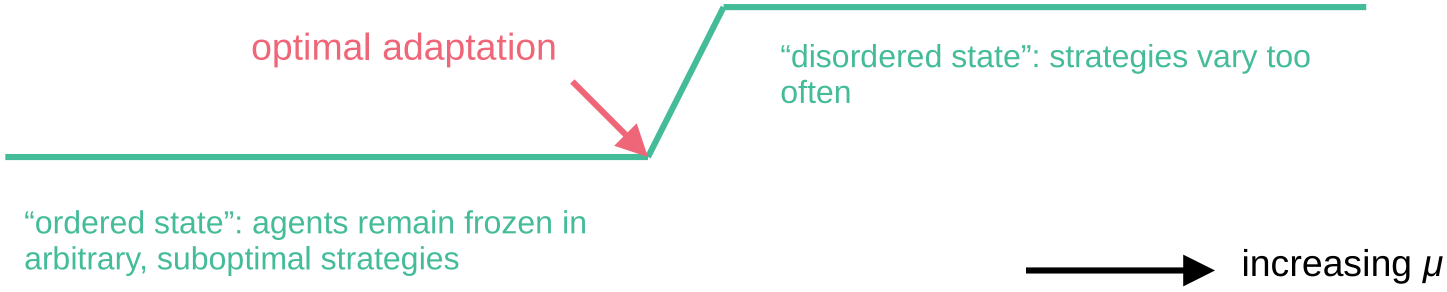
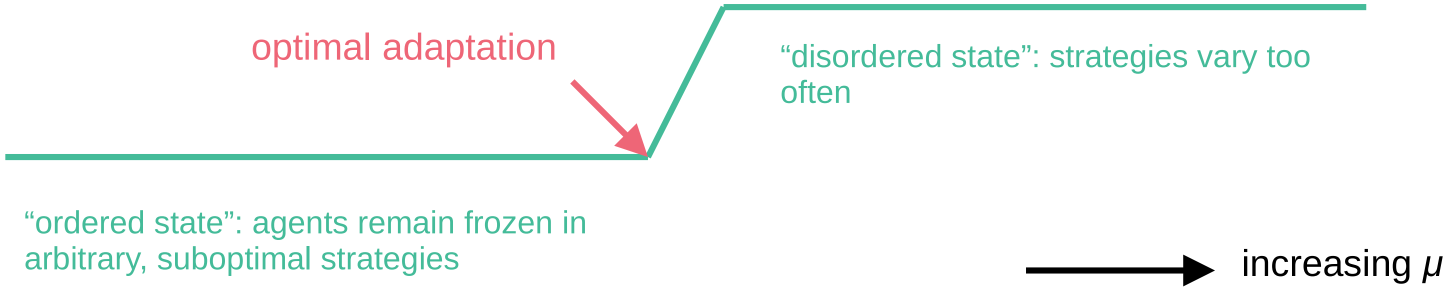


Weak emergence

Example 2: Packard’s model of evolving life

- Ordered and disordered macrostates (+ optimal mutation rate) are entirely derivable from initial microstates and system’s dynamics, but only by simulation



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  - → empirical support

- If this is emergence, then everything is emergent
  - → emergence as norm rather than exception

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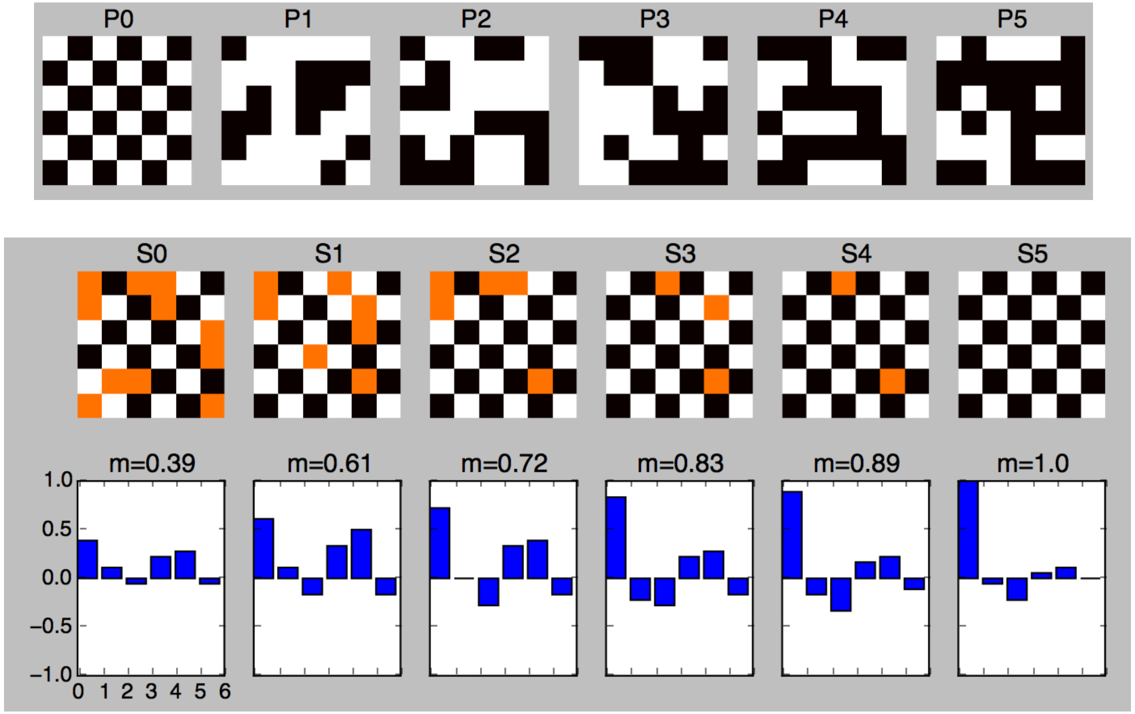
Discussion

- Weak emergence is compatible with the **causal closure of the physical** and the **exclusion principle**
- Also largely compatible with reductionism, taken in a certain sense
- No “real” downward causation: “downward causation in the weak form created by the activity of the micro-properties that constitute structural macro-properties”

The underdetermination of weak emergence

“Very weak emergence” in the Hopfield model



$$J_{ij} = \frac{1}{N}\sum_{\mu} p_i^{\mu} p_j^{\mu}$$ synaptic weights

update rule:

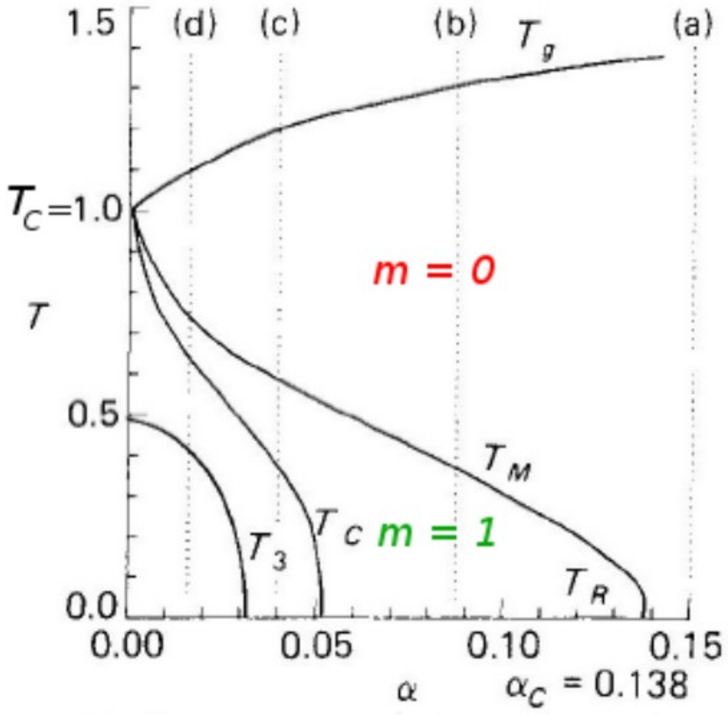
$$Pr(S_i(t) = 1) = \frac{1}{2}[1 + \tanh(\beta h_i(t-1) S_i(t-1))]$$

$$h_i(t) = \sum_j J_{ij} S_j(t)$$ local field

$m^{\mu}(t)$ quantifies the degree of similarity between the state of the network at time t and pattern μ

Images: https://neuronaldynamics-exercises.readthedocs.io/

The underdetermination of weak emergence



Phase diagram (Brunel, 2004)

- “Very weak emergence”: the system’s behavior can be derived by simulation, although there exists an analytical shortcut. However, such a derivation requires a mathematical idealization: $N \to +\infty$
- 2 possible bases for emergence:
  - Neurons + interactions
  - “memoryless” state
- Such a difference is not made in the literature

Outlook: towards a typology of emergent phenomena

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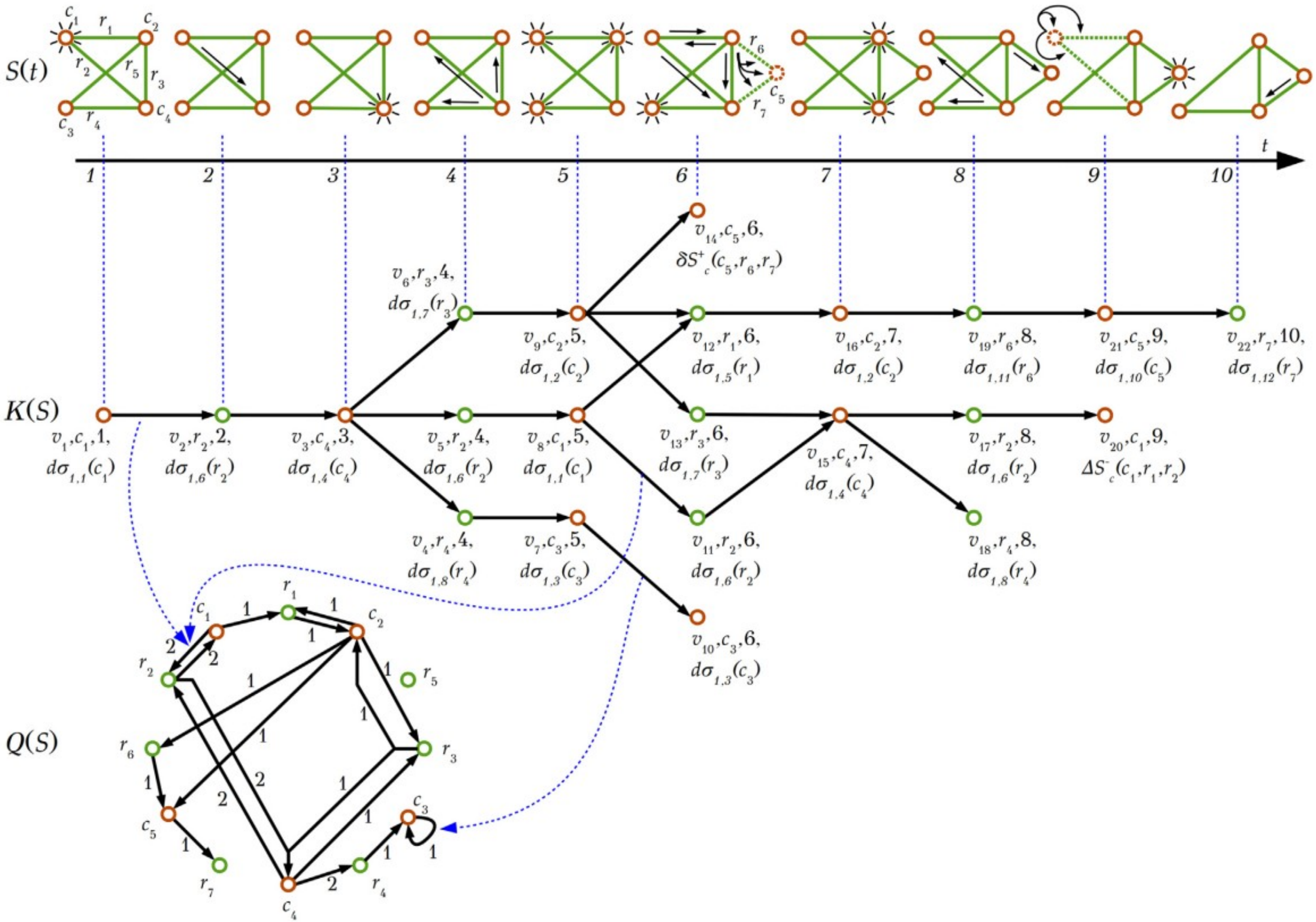
Original Research Article

Emergence and complex systems: The contribution of dynamic graph theory

Jacques Gignoux [a,*], Guillaume Chérel [b], Ian D. Davies [c], Shayne R. Flint [d], Eric Lateltin [a]

[a] *Institute of Ecology and Environmental Sciences, CNRS UMR 7618, 4 place Jussieu, bât. 44-45, CC 237, 75005 Paris, France*
[b] *Institut des Systèmes Complexes Paris Ile-de-France, 113 rue Nationale, 75013 Paris, France*
[c] *Fenner School of Environment and Society, The Australian National University, Canberra ACT 2601, Australia*
[d] *Research School of Computer Science, The Australian National University, Canberra ACT 0200, Australia*



S(t)
K(S)
Q(S)
t
1
2
3
4
5
6
7
8
9
10
$v_1, c_1, 1, d\sigma_{1,1}(c_1)$
$v_2, r_2, 2, d\sigma_{1,6}(r_2)$
$v_3, c_4, 3, d\sigma_{1,4}(c_4)$
$v_4, r_4, 4, d\sigma_{1,8}(r_4)$
$v_5, r_2, 4, d\sigma_{1,6}(r_2)$
$v_6, r_3, 4, d\sigma_{1,7}(r_3)$
$v_7, c_3, 5, d\sigma_{1,3}(c_3)$
$v_8, c_1, 5, d\sigma_{1,1}(c_1)$
$v_9, c_2, 5, d\sigma_{1,2}(c_2)$
$v_{10}, c_3, 6, d\sigma_{1,3}(c_3)$
$v_{11}, r_2, 6, d\sigma_{1,6}(r_2)$
$v_{12}, r_1, 6, d\sigma_{1,5}(r_1)$
$v_{13}, r_3, 6, d\sigma_{1,7}(r_3)$
$v_{14}, c_5, 6, \delta S^+_c(c_5, r_6, r_7)$
$v_{15}, c_4, 7, d\sigma_{1,4}(c_4)$
$v_{16}, c_2, 7, d\sigma_{1,2}(c_2)$
$v_{17}, r_2, 8, d\sigma_{1,6}(r_2)$
$v_{18}, r_4, 8, d\sigma_{1,8}(r_4)$
$v_{19}, r_6, 8, d\sigma_{1,11}(r_6)$
$v_{20}, c_1, 9, \Delta S^-_c(c_1, r_1, r_2)$
$v_{21}, c_5, 9, d\sigma_{1,10}(c_5)$
$v_{22}, r_7, 10, d\sigma_{1,12}(r_7)$

Outlook: towards a typology of emergent phenomena

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- **Discovery**: the system's behavior is unpredictable from the current models we have of it
- **Computational irreducibility (=weak emergence)**: to compute the system's macrostate, one needs to take all the system's details into account
- **Ontological emergence**: because its causal graph contains loops, the system becomes autonomous from external sources of change, and acquires new causal powers. Loops contain upward and downward causes.
  - *Is this form of emergence really observer-independent?*